BS 2 Linz | Class 4bITS



WEATHERLAB

IOT ENVIRONMENTAL MONITORING SYSTEM

Smart Weather Station & Quality Control

Continuous atmospheric sensing, local web server visualization, and live forecasting system.

PROJECT TEAM MEMBERS

David Sageder **Andrea Roithmeier** **Fabian Seelig** **Julian Weißböck** **Matthias Winklehner**

Project Assistance

David Steinegger

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Project roadmap & milestones.

Project Team & Roles

Clear task distribution among the 5 team members



Andrea Roithmeier

Project Lead

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David Sageder

Vice Project Lead & Hardware Engineer

Raspberry Pi setup.
OpenWeather API and data logging



Fabian Seelig

API & Backend Engineer

OpenWeather API and data logging.



Julian Weißböck

Documentation

Time tracking and weekly reports.



Matthias Winklehner

Web Developer

Project webpage and admin docs.

Project Management Standards

Structured execution, quality assurance, and weekly tracking



Organigram

Formal team structure and clear role descriptions for each member.

MANDATORY



CPM Network Plan

ProjectLibre plan identifying critical paths and milestones.

MANDATORY



Weekly Protocols

Signed weekly activity reports documenting progress and targets.

MANDATORY



Time Tracking

Kimai software logging personal workload and hourly effort.

MANDATORY

Project Roadmap & Milestones

Phased progression from initial setup to final presentation

COMPLETE TASK LIST (25 ITEMS)

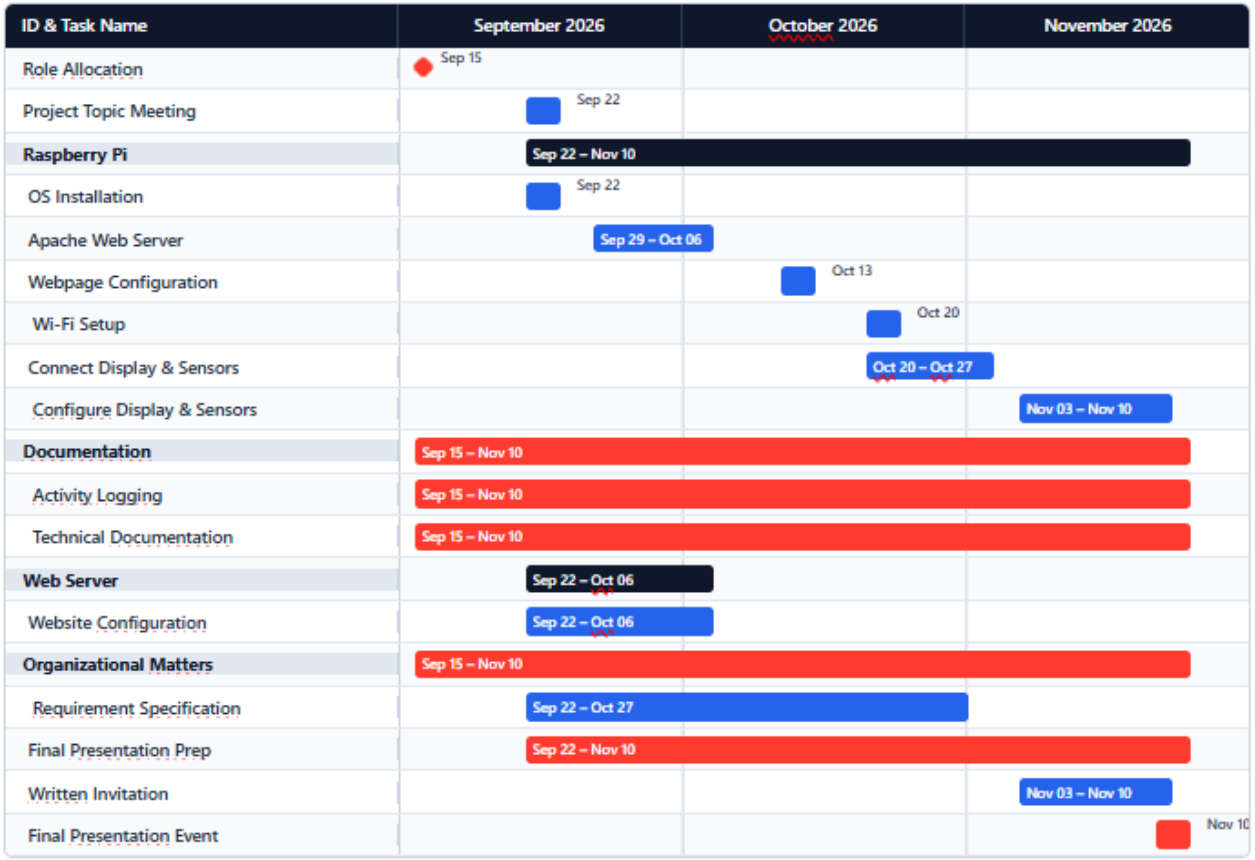
ID	Task Name	Duration	Start Date	Finish Date
	Role Allocation	1 day	15.09.26 08:00	15.09.26 17:00
	Project Topic Meeting	1 day	22.09.26 08:00	22.09.26 17:00
	Raspberry Pi	**8 days**	**22.09.26 08:00**	**10.11.26 17:00**
	↳ OS Installation	1 day	22.09.26 08:00	22.09.26 17:00
	↳ Apache Web Server	2 days	29.09.26 08:00	06.10.26 17:00
	↳ Webpage Configuration	1 day	13.10.26 08:00	13.10.26 17:00
	↳ Wi-Fi Setup	1 day	20.10.26 08:00	20.10.26 17:00
	↳ Connect Display & Sensors	1 day	20.10.26 17:00	27.10.26 17:00
	↳ Configure Display & Sensors	2 days	03.11.26 08:00	10.11.26 17:00
	Documentation	**9 days**	**15.09.26 08:00**	**10.11.26 17:00**
	↳ Activity Logging	9 days	15.09.26 08:00	10.11.26 17:00
	↳ Technical Documentation	9 days	15.09.26 08:00	10.11.26 17:00
	Web Server	**3 days**	**22.09.26 08:00**	**06.10.26 17:00**
	↳ Set up Linux VM	1 day	22.09.26 08:00	22.09.26 17:00
	↳ Network Configuration	1 day	22.09.26 08:00	22.09.26 17:00
	↳ Apache Server Installation	1 day	22.09.26 08:00	22.09.26 17:00
	↳ Website Configuration	3 days	22.09.26 08:00	06.10.26 17:00
	Organizational Matters	**9 days**	**15.09.26 08:00**	**10.11.26 17:00**
	↳ Project Charter / Spec Sheet	1 day	15.09.26 08:00	15.09.26 17:00
	↳ Intro Presentation	1 day	29.09.26 08:00	29.09.26 17:00
	↳ Detailed Requirement Spec	6 days	22.09.26 08:00	27.10.26 17:00
	↳ Final Presentation Prep	8 days	22.09.26 08:00	10.11.26 17:00
	↳ Written Invitation	1 day	03.11.26 09:00	10.11.26 09:00
	↳ Topic Presentation	1 day	29.09.26 08:00	29.09.26 17:00
	↳ Final Presentation Event	0.87 days	10.11.26 09:00	10.11.26 17:00

WeatherLab – Project Timeline

Complete overview of all 25 project tasks

TOTAL TASKS	MAIN GROUPS	PROJECT START	PROJECT FINISH
25 Tasks	5 Packages	Sep 15, 2026	Nov 10, 2026

VISUALIZED PROJECT SCHEDULE (SEP – NOV 2026)



What is Our Vision?

Compact environmental monitoring with local touch controls

Primary Goal

Build a self-contained IoT weather station providing live indoor air quality readings and outdoor forecast projections.

- **BME680 Sensor Integration**
- **Local Touch-Display GUI**
- **OpenWeather API 3-Day Forecast**

Live Environmental Data

Continuous tracking of room temperature, humidity, pressure, and air quality.

Integrated Web Server

Raspberry Pi hosts local Web-GUI accessible directly on the 7-inch touch screen.

Documentation Web Portal

Dedicated Virtual Machine hosting admin setup guides and smart home features.

Smart Home & Future Outlook

Expansions and advanced automation integrations



Telegram Bot Warnings

Automated instant notifications for severe weather alerts, storm warnings, or sudden temperature drops.

CAN Feature



Air Quality LED Feedback

Physical LED indicator changing color based on BME680 indoor air quality index readings.

CAN Feature



Smart Home Hub

Centralized control platform for additional home automation devices and remote web access.

Smart Home Integration

Technical Requirements

Breakdown of mandatory core features and optional enhancements

MUST Requirements

MANDATORY

- **Sensor Hardware Hookup:**
  BME680 wired to Raspberry Pi for continuous reading.
- **Local Web-GUI & Display:**
  Live presentation on 7-inch touch screen via local server.
- **External API Integration:**
  Fetch 3-day weather forecast using OpenWeather API.
- **VM Project Portal:**
  Documentation, hardware introduction, and admin guides.

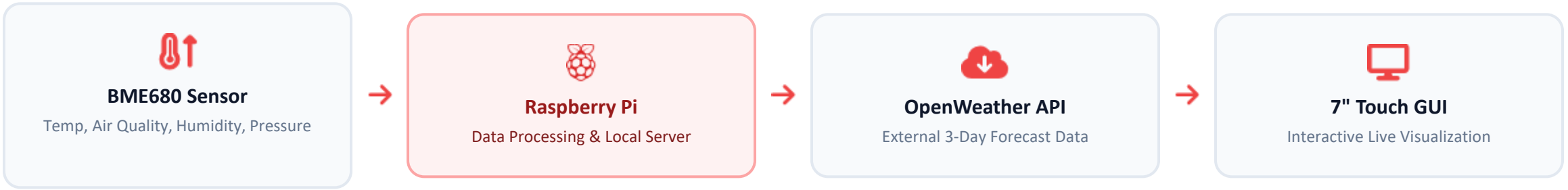
CAN Requirements

OPTIONAL

- **Telegram Alerts:**
  Automated storm and extreme temperature warnings.
- **Smartphone Webview**
  A Webview of the Forecast Interface for Smartphones
- **Visual LED Indicator:**
  Physical air quality light feedback signal.

System Architecture & Flow

Data pipeline from hardware sensors to touchscreen presentation



Continuous Interval Logging

Sensors read environment at fixed intervals and store values locally for analysis.

Documentation Virtual Machine

Independent VM web server serving administrator setup guides and system manuals.

Cost

Material and personnel costs of the WeatherLab project

Item	Cost
Raspberry Pi (provided)	€0.00
BME680 sensor	€26.29
7-inch touch monitor	€65.87
microSD card 256 GB (estimated)	€30.00
Power supply & cables (estimated)	€15.00
Software & services (OpenWeather Free, ProjectLibre, Teams)	€0.00
Total material	**€137.16**
Personnel (5 people × 30 h × €20)	€3,000.00
Total cost	**€3,137.16**

MATERIAL COSTS

€137.16

PERSONNEL COSTS (FICTIONAL)

€3,000.00

TOTAL PROJECT COST

€3,137.16



WEATHERLAB

Thank You for Your Attention!

Questions & Discussion

Project Team
David S. | Andrea R. | Fabian S. | Julian W. | Matthias W.

Project Supervision
BS 2 Linz - Steinegger David